ME-OPS v2 — Technical Handoff Dossier

Forensic-Grade Inheritance Package

Owner: Johnny Cage (MPLSJohnny Cage) | **Machine:** GettUppENT (Windows 11, user: finan) | **Date:** 2026-03-21

EXECUTIVE SUMMARY

ME-OPS v2 is a **Personal Causal Intelligence System** that sits on top of PiecesOS and converts raw digital activity history into reusable failure chains, win signatures, heuristics, and operator briefings. It is NOT a memory app. It is a causal operating system over lived digital evidence.

Status as of 2026-03-21:

- Phase 1 (ledger + ingestion): ✅ Complete
- Phase 2 (cases + sessions + failures): ✅ Complete
- Phase 3 (Gemini semantic recall): ✅ Live — 579 embeddings in DB, cosine search working
- Phase 4 (heuristic mining): ⬜ Not started
- Phase 5 (live operator guardrails): ⬜ Not started

The system is operational today. You can run `python run_phase2.py recall <query>` and get semantically ranked prior cases from your actual history.

MVP DEFINITION

Goal: A system that answers these 8 questions from evidence:

1. What exact sequence usually precedes a win?
2. What exact sequence usually precedes wasted hours?
3. What environments repeatedly cause drag?
4. Which decisions were high-confidence but wrong?
5. Which tools produce real leverage vs fake progress?
6. What should I do first when this kind of problem appears?
7. Which anti-pattern am I repeating right now?
8. What is the best known next move based on evidence?

MVP scope is Phase 1–3. Phases 4–5 are intelligence amplification on top of a working foundation.

Out of scope for MVP: GUI/dashboard, autonomous agent integration, multi-user support, cloud sync, real-time streaming ingestion via WebSocket (polling fallback is in place).

INTENDED ARCHITECTURE vs CURRENT REALITY

Intended Stack

```
PiecesOS (localhost:39300)
    └── WebSocket stream / REST API
         └── ME-OPS Ingestion Layer
              └── Canonical Ledger (SQLite)
                   ├── Cases (failure chains, win signatures)
                   ├── Heuristics (operating laws)
```

```
                    ├── Embeddings (Gemini 768d vectors)
                    └── Operator Briefings
```

Actual Implementation

- **WebSocket streaming:** Stubbed. Falls back to REST polling via `mcp_client.py`. The WebSocket path exists in `ingest.py` (original) but `ingest_p2.py` (active) uses REST + local cache re-classification.
- **MCP protocol bridge:** Working. Uses SSE `http://localhost:39300/model_context_protocol/2024-11-05/sse` to batch-fetch summaries via `workstream_summaries_batch_snapshot`.
- **Canonical ledger:** Working. 526 events in SQLite.
- **Cases:** Working but thin. 9 failure chains + 9 win signatures from 26 failures and 9 outcomes. Detection signals needed expansion (was done in classify.py).
- **Embeddings:** Working. 579 entities embedded at 768d via `gemini-embedding-2-preview`. Free tier: 100 RPM, batch size 50, 65s inter-batch delay.
- **Heuristics:** 10 manually seeded. Auto-mining (Phase 4) not yet built.
- **Briefings:** Template-based, not ML-driven. Hardcoded narratives with live DB stat injection.
- **Decisions:** 0 extracted. Decision trigger patterns exist in `decision_replay.py` but no events matched. Likely needs tuning to workstream summary language.
- **Context collapse detector:** Built but produces 0 alerts. Sessions have no fragmentation score > 0.6 under current scoring.

FILE AND FOLDER INVENTORY

Root files

File	Status	Purpose	Safe to modify?
`run.py`	Active (Phase 1 only)	Original Phase 1 runner. Calls `run.py stats`, `sync`, `cases`, `brief`, `seed`	Yes — superseded by run_phase2.py
`run_phase2.py`	**Primary entry point**	Full Phase 2+3 pipeline runner. All commands route through here	Yes — main CLI interface
`requirements.txt`	Active	`pydantic>=2.0`, `websockets>=12.0` — NOTE: missing `google-genai` and `numpy`	**Must update**
`meops.db`	**Critical**	Live SQLite database. All canonical data lives here. Not in git.	Never delete without backup
`_debug.py`	Scratch	One-off diagnostic. Shows annotation_text sample. Can delete.	Delete
`_enc.py`	Scratch	Encoding test. Can delete.	Delete
`_migrate.py`	Completed	Added `fragmentation_score` to `context_metrics`. Already ran.	Delete
`_reset_canonical.py`	Dangerous	Wipes all derived layers, resets raw cache. Use only to rebuild from scratch.	Keep but label clearly
`_test_embed.py`	Scratch	Verifies Gemini API key and embedding model work.	Keep as health check

core/models/

File	Status	What it models
`base.py`	Active	BaseRecord, ObjectType enum, SourceKind, ActorKind
`event.py`	Active	Event, Artifact, SystemState with EventKind/LocationKind enums
`decision.py`	Active	Decision (with quality scoring formula), Failure, Outcome
`case.py`	Active	Case, Heuristic, Briefing, Intervention models

Note: Pydantic models are defined but the DB uses raw JSON blobs via `EntityRepo.upsert()`. The models are not validated on write — they serve as documentation and reference. This is a known gap.

core/storage/

File	Status	Purpose
`schema.py`	**Critical**	Full SQLite schema SQL + SEED_PROJECTS_SQL. Source of truth for DB structure.
`db.py`	**Critical**	`get_conn()`, `init_db()`, `EntityRepo` (upsert/get/find_by_type/add_edge), `ensure_project()`

Known issue: `schema.py` `context_metrics` table was missing `fragmentation_score` column at creation. Fixed via `_migrate.py` ALTER TABLE. If you `init_db()` fresh, the column now exists in SCHEMA_SQL — this is corrected.

core/ledger/

File	Status	Purpose
`classify.py`	**Critical**	Maps raw text → project_id and event_kind. Rule-based keyword matching. 13 projects seeded.
`sessionize.py`	Active	Groups events into sessions by 30-min gap / 4-hour max. Computes fragmentation/context_load scores.

Known gap: `classify.py` leaves ~311/526 events as `project_id="unknown"`. The unknown project is seeded in DB as a catch-all. These are predominantly WSL/IDE infrastructure sessions that don't match any venture keyword. Add more signals or use Gemini classification to improve.

core/cases/

File	Status	Purpose
`failure_chain.py`	Active	`lift_failures_and_outcomes()` + `build_failure_chains()`. Scans events for failure/outcome signals, clusters into chains.
`win_signature.py`	Active	`build_win_signatures()`. Scans outcome entities, builds win cases with preconditions/sequences.
`decision_replay.py`	Active but 0 output	`extract_decisions()`. Scans for 30+ decision trigger phrases. Currently extracting 0 decisions — triggers don't match workstream summary language well enough.

core/recall/

File	Status	Purpose
`gemini_embed.py`	**Active — Phase 3**	Full Gemini embedding pipeline. Batch embed, store as SQLite blob, cosine search, format recall report. Rate limit: 100 RPM free tier, batch 50, 65s delay, respects retryDelay from 429.
`retrieve.py`	Active — TF-IDF fallback	TFIDFIndex class with cosine sim. Used when Gemini quota is exceeded.

core/heuristics/

File	Status	Purpose
`seed.py`	Active	10 manually seeded heuristics derived from observed patterns. Auto-mining (Phase 4) not built.

core/briefing/

File	Status	Purpose
`daily.py`	Active	3 briefing templates (daily, incident, project). Template-based — not ML. Uses live DB counts + hardcoded narratives.

core/operator/

File	Status	Purpose
`collapse_detector.py`	Active but 0 output	Scans sessions, scores collapse risk, writes interventions. Currently produces 0 alerts — sessions score below 0.6 threshold because tool_switch_count comes from session tool set size, not actual switch events.

core/causal/

File	Status	Purpose
(empty, `__init__.py` only)	**Stubbed**	Phase 4 causal inference module. Not built.

pieces_bridge/

File	Status	Purpose
`ingest.py`	**Superseded**	Original bridge using REST `/workstream_summaries` endpoint — returns 404 on all IDs. Dead.
`ingest_p2.py`	**Active**	Working bridge: re-classifies 526 events from local raw cache, fetches delta from MCP batch_snapshot.
`mcp_client.py`	**Active**	Pure MCP protocol over SSE. `fetch_all_summary_ids_mcp()` + `fetch_summaries_batch_mcp()`.

Critical: `ingest.py` is dead. The REST endpoint `/workstream_summaries/{id}` returns 404 for all IDs. Only the MCP path works. Do not attempt to use `ingest.py`.

pipelines/

File	Status	Purpose
`__init__.py` only	**Empty scaffold**	Intended for standalone pipeline scripts. Not used.

VERSIONS AND DEPENDENCY MATRIX

Component	Version	Location	Notes
Python	3.14.2	`C:\Python314\`	Newest major. Some pip packages may not support 3.14 yet.
pip	25.3	Global	Upgrade available (26.0.1) — not required
google-genai	1.68.0	User site-packages	SDK for Gemini API
pydantic	2.12.0	User site-packages	Data models
numpy	2.4.3	User site-packages	Vector math for cosine sim
websockets	16.0	User site-packages	Required by google-genai; also used in WS bridge
node.js	v22.22.1 (via nvm)	`C:\Users\finan\AppData\Local\nvm\`	Used by MCP test scripts
mcp-remote	0.1.38	npm global	MCP bridge for Claude Desktop
PiecesOS	12.3.10	Windows system tray	Must be running for any ingestion
SQLite	built-in Python	stdlib	No separate install needed

Missing from requirements.txt: `google-genai`, `numpy`. Must add before fresh install on another machine.

Correct requirements.txt should be:

```
pydantic>=2.0
google-genai>=1.68.0
numpy>=2.0
websockets>=12.0
```

MACHINE AND ENVIRONMENT PROFILE

- **OS:** Windows 11 (user: `finan`, machine name: `GettUppENT`)
- **Shell:** PowerShell (default) + cmd.exe for some commands
- **Python location:** `C:\Python314\python.exe`

- **Site packages:** `C:\Users\finan\AppData\Roaming\Python\Python314\site-packages\`
- **DB location:** `C:\Users\finan\me-ops\meops.db` (hardcoded via `MEOPS_DB` env var or default `Path.home() / "me-ops" / "meops.db"`)
- **PiecesOS port:** 39300 (hardcoded in `mcp_client.py` and `gemini_embed.py`)
- **Gemini API key:** `AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes` — **HARDCODED in run_phase2.py and gemini_embed.py**. Move to env var before sharing.
- **PYTHONIOENCODING:** Must be set to `utf-8` on Windows. `run_phase2.py` sets it at startup. All other scripts are at risk of `charmap` errors on `→` , ✅ , emoji characters.
- **WSL:** Present (WSL2, Ubuntu). Not used by ME-OPS directly. Used by Paperclip/OpenClaw stack.
- **No Docker, no containers, no CI/CD, no .env file.**
- **No test suite.** Zero test coverage.

Machine-specific hacks

- `"unknown"` project was manually inserted into the projects table mid-session to unblock FK constraint failures. It is now in SEED_PROJECTS_SQL so fresh installs will have it.
- `context_metrics.fragmentation_score` column was added via ALTER TABLE after schema was deployed. Fixed in schema.py but if you restore an old DB snapshot, run `_migrate.py` first.

SETUP AND RECOVERY RUNBOOK

Cold Start (fresh machine)

```
# 1. Prerequisites
# Install Python 3.14+ (https://python.org)
# Install PiecesOS and enable LTM-2.7 engine
# Set GEMINI_API_KEY env var

# 2. Clone / copy project
cd C:\Users\{you}

# 3. Install dependencies
pip install pydantic>=2.0 google-genai>=1.68.0 numpy>=2.0 websockets>=12.0

# 4. Initialize DB and seed heuristics
cd me-ops
set PYTHONIOENCODING=utf-8
set GEMINI_API_KEY=YOUR_KEY_HERE
python run_phase2.py full

# 5. Verify
python run_phase2.py stats
python run_phase2.py recall "IronClad revenue blocked"
```

Resume Existing Install (embed-only)

```
cd C:\Users\finan\me-ops
set PYTHONIOENCODING=utf-8
```

```
set GEMINI_API_KEY=AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes
python run_phase2.py embed
```

Full Reset (wipe derived layers, re-run everything)

```
python _reset_canonical.py     # wipes entities/cases/sessions, resets raw cache
python run_phase2.py full       # re-ingests + rebuilds + re-embeds
```

Health Check

```
python _test_embed.py           # verifies Gemini API key
curl http://localhost:39300/.well-known/version   # verifies PiecesOS
python run_phase2.py stats      # shows DB counts
```

ERROR AND FAILURE HISTORY

Error	Where	Root Cause	Status	Blocks MVP?
`HTTP 404` on all `/workstream_summaries/{id}` REST calls	`ingest.py`	Wrong REST endpoint path	**Fixed** — replaced by MCP path in `ingest_p2.py`	Was blockin
`sqlite3.IntegrityError: FOREIGN KEY constraint failed`	`failure_chain.py`, `win_signature.py`, `decision_replay.py`	`project_id="unknown"` not in projects table	**Fixed** — `unknown` project seeded, `_safe_pid()` added	Was blockin
`UnicodeEncodeError: 'charmap' codec`	`run_phase2.py` (and any script using → ✅)	Windows default cp1252 encoding	**Fixed** — `PYTHONIOENCODING=utf-8` set at top of run scripts	Low severity
`sqlite3.OperationalError: table context_metrics has no column fragmentation_score`	`collapse_detector.py`	Schema deployed before column existed	**Fixed** — column in SCHEMA_SQL now, `_migrate.py` ran on existing DB	Was blockin
`AttributeError: 'sqlite3.Row' has no attribute 'get'`	`collapse_detector.py`, `win_signature.py`	`sqlite3.Row` doesn't support `.get()` — use `row["key"]`	**Fixed** in both files	Was blockin
`429 RESOURCE_EXHAUSTED` on Gemini embed	`gemini_embed.py`	Free tier 100 RPM limit hit after first batch of 100	**Fixed** — batch size 50, 65s delay, `retryDelay` extracted from error and honored	Recurrin — expecte

`SyntaxError: invalid syntax` in interactive Python	Multiple	PowerShell interprets `|` in cmd strings as pipe	Workaround: use `.py` files instead of `-c` inline scripts	Low
`ERR_UNESCAPED_CHARACTERS` in Node.js MCP test	`pieces_mcp_test.js`	Token in SSE URL contains `=` chars, passed as raw string	Fixed in `pieces_ltm_call.js` with `new URL()` parsing	Fixed
`0 decisions extracted`	`decision_replay.py`	Decision trigger phrases don't match workstream summary narrative style	**Unresolved** — needs better trigger set or embedding-based detection	Not MVP-blockin
`0 collapse alerts`	`collapse_detector.py`	tool_switch_count computed from distinct tools per session, not actual switch events	**Unresolved** — needs real switching signal from clipboard/app events	Not MVP-blockin

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DATABASE SCHEMA STATE

Location: `C:\Users\finan\me-ops\meops.db`

Tables and row counts (2026-03-21):

Table	Rows	Notes
entities	579	526 events + 26 failures + 9 outcomes + 0 decisions + 0 interventions
projects	13	All ventures + wsl-infra, blink-agent, claude-integration, unknown
sessions	207	Grouped by 30-min gap / 4-hr max
cases	18	9 failure chains + 9 win signatures
case_members	27	Links cases to entities
heuristics	10	All manually seeded
edges	44	TRIGGERED + RESULTED_IN + DERIVED_FROM
embeddings	579	All entities + cases at 768d float32 blobs
raw_pieces_summaries	1374	Raw cache. 526 processed=1. 848 unprocessed (no annotation text or bad fetch).
raw_import_batches	Multiple	Each run creates one batch row
sessions	207	
context_metrics	207	All sessions logged but collapse_score = 0 for all

briefings	0	Not written — briefings only print, not persisted
interventions	0	Collapse detector produces 0 alerts

Schema integrity: FOREIGN KEY constraints active. All project_ids resolve. No orphaned records known.

Trustworthy for Phase 4: Yes — the canonical layer is stable enough to mine heuristics from.

CONFIRMED WORKING / PARTIAL / BROKEN / STUBBED

Confirmed Working ✅

- Full pipeline: `python run_phase2.py full`
- DB init + schema + seed
- MCP ingestion from PiecesOS via SSE batch_snapshot
- Local cache re-classification (525 events from 1374 raw)
- Sessionization (207 sessions)
- Failure lifting (26 failures from 526 events)
- Outcome lifting (9 outcomes)
- Failure chain building (9 chains)
- Win signature building (9 sigs)
- Gemini embedding (579 at 768d, rate-limited correctly)
- Cosine semantic recall (`run_phase2.py recall <query>`) — confirmed 0.74–0.76 similarity on IronClad queries
- TF-IDF fallback recall when Gemini quota exceeded
- Daily operator briefing output
- Project briefing output
- 10 seeded heuristics

Partial / Needs Improvement ⚠️

- **Project classification:** 311/526 (59%) unclassified as "unknown". Core ventures well-covered (79 IronClad, 36 wsl-infra, 33 openclaw, 28 antigravity). Unknown pool is legitimate infrastructure sessions without venture-specific keywords.
- **Failure detection:** Only 26/526 events flagged as failures. Broadened detection signals help but real failure density in workstream summaries is lower than raw error logs.
- **Context collapse detection:** 0 alerts despite known session fragmentation. Scoring logic needs real app-switch data from workstream events, not just session tool counts.

Broken / Dead ❌

- `ingest.py` (original bridge) — REST endpoint returns 404 for all summary IDs. Do not use.
- `decision_replay.py` — 0 decisions extracted from 526 events. Trigger phrases don't match.
- `pipelines/` folder — completely empty scaffold. Not wired to anything.
- `core/causal/` — empty. Phase 4 inference not built.
- WebSocket real-time streaming — code path exists in `ingest.py` but that file is dead. `ingest_p2.py` uses polling.

Stubbed / Placeholder ⬜

- `core/briefing/daily.py` — briefing content is hardcoded narrative, not derived from actual case analysis. Works as template but is not intelligent yet.
- `core/models/*.py` — Pydantic models defined but not enforced on DB writes. Reference only.

RISKS AND SECURITY CONCERNS

Risk	Severity	Notes
Gemini API key hardcoded in `run_phase2.py` and `gemini_embed.py`	🔴 HIGH	`AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes` visible in plaintext. Move to `.env` or env var immediately before sharing repo.
No `.env` file	🔴 HIGH	No secrets management at all. Key lives in source.
No backups of `meops.db`	🟡 MEDIUM	Single SQLite file. One accidental delete or corruption = all data lost. Schedule backup or copy before any schema migration.
`_reset_canonical.py` is destructive	🟡 MEDIUM	Wipes all derived entities. No confirmation prompt. Mislabeled as "reset" not "destroy".
No test suite	🟡 MEDIUM	Zero test coverage across all modules. Any refactor risks silent regression.
Port 39300 hardcoded	🟡 MEDIUM	PiecesOS port varies per install. Should read from PiecesOS Quick Menu config.
Free tier Gemini quota	🟢 LOW	100 RPM limit. System handles gracefully with retry logic. Upgrade to paid tier for bulk re-embedding.
Machine coupling	🟢 LOW	DB path defaults to `C:\Users\finan\me-ops\meops.db`. Portable via `MEOPS_DB` env var but not documented.

MAINTENANCE GUIDANCE

Where to start

1. `python run_phase2.py stats` — current DB state
2. `python run_phase2.py recall <query>` — verify recall is working
3. Read `core/ledger/classify.py` — most impactful file for data quality

What NOT to change first

- `core/storage/schema.py` — any table changes require migration + data wipe
- `core/storage/db.py` — `EntityRepo.upsert()` is load-bearing for all ingestion
- `meops.db` — never edit directly; always through Python

High-risk modules

- `core/storage/schema.py` — schema changes break existing DB
- `pieces_bridge/mcp_client.py` — depends on exact PiecesOS MCP protocol version path
- `core/recall/gemini_embed.py` — API key, rate limits, blob format

Safe to iterate freely

- `core/ledger/classify.py` — adding keywords only improves classification
- `core/heuristics/seed.py` — add more seeded heuristics
- `core/briefing/daily.py` — improve templates

- `run_phase2.py` — add new commands

Before any schema change

1. `copy C:\Users\finan\me-ops\meops.db C:\Users\finan\me-ops\meops.db.bak`
2. Write migration as ALTER TABLE statements in a `_migrate_vN.py` file
3. Test on fresh DB first: `set MEOPS_DB=C:\Users\finan\me-ops\meops_test.db && python run_phase2.py full`

UPGRADE ROADMAP

Phase	What	Effort	Priority
4	Heuristic mining — contrastive win/failure analysis, frequency mining, auto-generate operating laws	Medium	High
4.5	Fix decision extractor — embed-based detection instead of keyword triggers	Low	Medium
4.5	Fix context collapse — use workstream_events (clipboard/vision/audio) for real app-switch signal	Medium	Medium
5	Live operator guardrails — dead-end alert, context collapse interrupt, best-next-move	Medium	High
5.5	`.env` file + secrets management	Low	**Do immediately**
5.5	Test suite — at least classify.py, failure_chain.py, gemini_embed.py	Medium	High before Phase 5
6	OpenClaw integration — ME-OPS as MCP tool for autonomous daily briefing	Medium	Post-MVP
6	Paperclip CEO agent integration — ME-OPS feeds governance layer	High	Post-MVP

IMMEDIATE NEXT ACTIONS (PRIORITY ORDER)

1. **SECURITY:** Move Gemini API key to `.env` file. Add `python-dotenv` to requirements.
2. **BACKUP:** `copy meops.db meops.db.bak.20260321`
3. **FIX requirements.txt:** Add `google-genai>=1.68.0` and `numpy>=2.0`
4. **DELETE scratch files:** `_debug.py`, `_enc.py`, `_migrate.py`
5. **CLASSIFY unknowns:** Add ~20 more keyword signals to `classify.py` for the 311 unknown events — look at their `summary` field in DB to find patterns
6. **PHASE 4:** Build `core/heuristics/mine_rules.py` — frequency mining on failure chains, contrastive analysis of win vs failure sequences
7. **FIX decisions:** Replace keyword triggers in `decision_replay.py` with embedding similarity search against known decision phrases
8. **ADD test:** At minimum one test per module before Phase 5

OPEN QUESTIONS

1. **Unknown project classification:** What are those 311 events? Run `SELECT summary FROM entities WHERE project_id='unknown' LIMIT 20` to inspect — likely they're general AI tooling research sessions that need new project buckets (e.g. `ai-research`, `personal`).
2. **Paperclip + OpenClaw:** Should ME-OPS feed into the Paperclip CEO agent? The clipboard captured a full CEO system prompt for Paperclip — integration design needed.
3. **ME-OPS v2 spec:** The full architecture spec is in clipboard history (captured by PiecesOS). It includes 20 sections of canonical schema, pipeline design, and scoring formulas. The current implementation covers ~40% of that spec.
4. **Paid Gemini tier:** 100 RPM free limit means full re-embedding of 526 entities takes ~11 minutes. Paid tier removes this constraint.
5. **Airbnb cleaning venture:** A complete GTM for a Minneapolis STR cleaning service was generated in a single session (March 20) and never acted on. Captured in clipboard. Worth reviewing.

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